

OpenSpace Fintech WhatsApp Bot

Company: OpenSpace
Stack: Node.js • Baileys • Groq LLM
Production Ready

System Design, Architecture & Workflow Documentation

1. Executive Summary

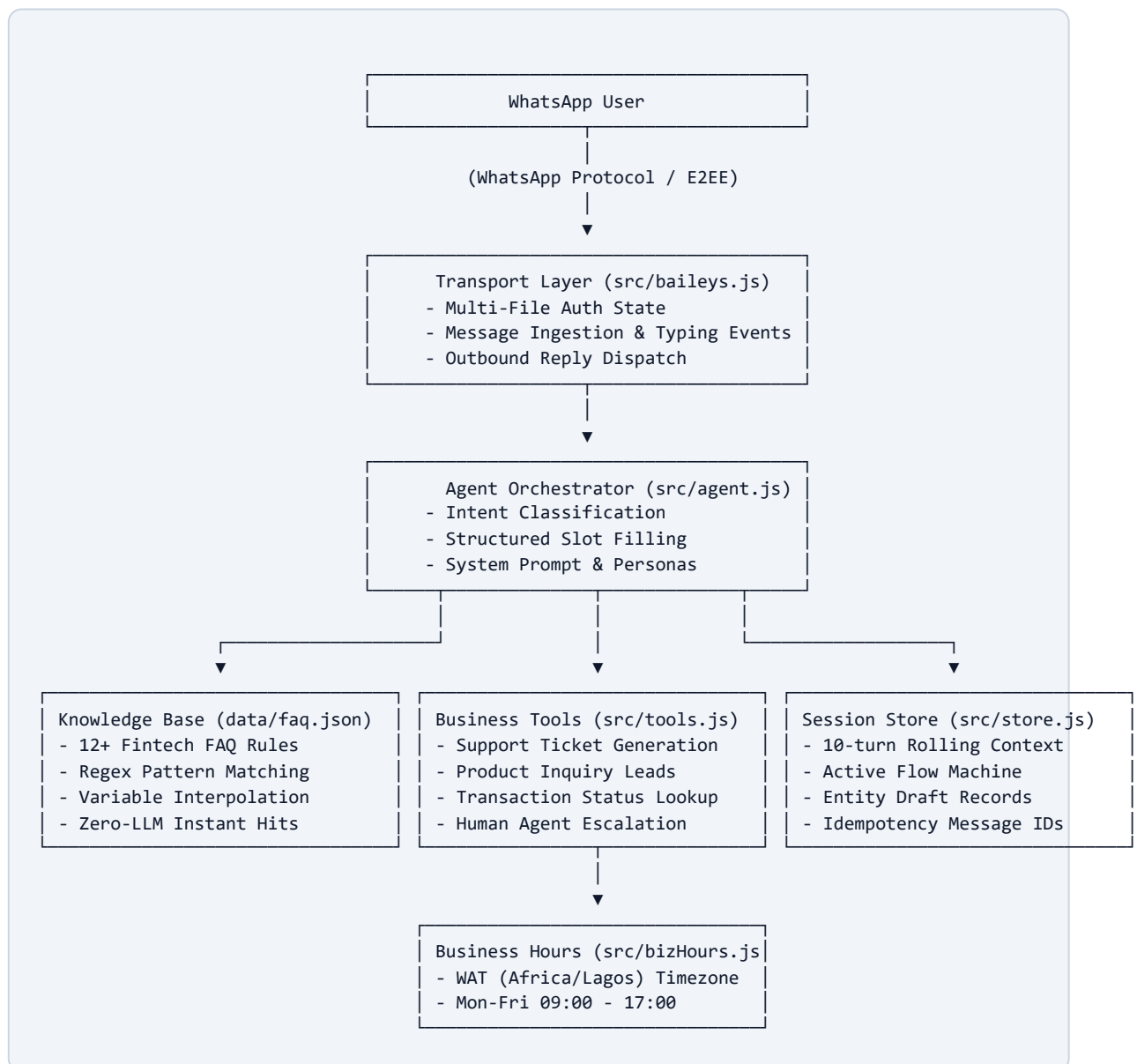
The **OpenSpace WhatsApp AI Assistant** is an enterprise-grade automated agent designed to provide instant, 24/7 financial support, onboarding assistance, loan intake, agent banking guidance, and dispute management for **OpenSpace** customers via WhatsApp.

By pairing **Baileys (direct WhatsApp Web multi-device WebSocket protocol)** with high-performance LLM inference via the **Groq API**, the bot delivers sub-second response times with zero monthly API markup fees, strict slot-filling state machines, and complete data privacy.

2. Technology Stack

Layer	Technology	Description / Role
Runtime	Node.js (ES Modules)	Asynchronous, non-blocking runtime handling WebSockets and AI requests.
WhatsApp Connector	@whiskeysockets/baileys	Lightweight, browserless WhatsApp Web protocol engine with multi-device persistence.
AI / LLM Engine	Groq Cloud (openai/gpt-oss-120b)	High-speed open-source Large Language Model for intent classification & JSON extraction.
AI Client Interface	openai SDK	Universal client interface configured with Groq base URL for low-latency inference.
Knowledge Engine	Deterministic Regex & Template Engine	Pre-compiled regex pattern matching in <code>faq.json</code> with dynamic environment replacement.
Session Memory	In-Memory Rolling State	Tracks 10-turn message history, active multi-turn flows (DISPUTE, INQUIRY), and entity drafts.
Timezone / Hours	Intl.DateTimeFormat	West Africa Time (Africa/Lagos) business hours validator (Mon–Fri 09:00–17:00).

3. High-Level Architecture



4. Core Product Workflows & Capabilities

Account & Wallet Creation

Guides users through fast onboarding, BVN/NIN/ID KYC verification, and instant virtual account creation.

Business Banking

Handles inquiries for corporate accounts, invoicing tools, payroll automation, and merchant APIs.

Personal & Business Loans

Interactive intake capturing loan amount, business type, and applicant info to issue a REQ-xxxxxx lead.

Open Nearby (Agency Banking)

Explains cash-in/cash-out agency points, POS deployment, and agent onboarding steps.

Open Invest

Details high-yield savings plans, locked deposits, and structured investment yields.

Dispute & Issue Resolution

Guided slot filling for failed transfers or double debits, generating a trackable TKT-xxxxxx ticket.

5. Conversational Flow & State Machine

The assistant employs a multi-tiered routing process to resolve user intent:

- 1. Direct Pattern Matching:** Incoming queries are evaluated against `data/faq.json` regexes. If a direct match is found (e.g., support hours, contact info), the templated answer is returned instantly without querying the LLM.
- 2. Intent Classification (LLM):** Unmatched queries are processed by the LLM in structured JSON mode to categorize user intent:
 - `DISPUTE` $\rightarrow$ Enters multi-turn dispute flow.
 - `INQUIRY` $\rightarrow$ Enters multi-turn loan/product application flow.
 - `TX_STATUS` $\rightarrow$ Executes transaction status stub.
 - `HANDOFF` $\rightarrow$ Checks business hours and escalates to human agents.
 - `GENERAL` $\rightarrow$ Returns conversational AI answer.
- 3. Slot-Filling Guided State Machine:** If inside a flow, the agent prompts for missing fields one-by-one (Amount, Reference, Name, Email) and confirms completion with an official reference ID.

6. Security, Compliance & Deployment

- **Zero Hardcoded Credentials:** All keys and endpoints are configured exclusively in `.env`.
- **Private Auth State:** The WhatsApp session directory (`auth_info_baileys/`) is excluded from version control in `.gitignore`.

- **Operating Schedule:** Built-in awareness of West Africa Time (WAT) operating hours with automatic offline escalation.

7. Environment Variables Reference

```
# AI Inference (Groq Cloud)
GROQ_API_KEY=gsk_XXXXXXXXXXXXXXXXXXXXXXXXXXXXX
GROQ_MODEL=openai/gpt-oss-120b

# WhatsApp Pairing (Optional for Code Pairing)
BOT_PHONE_NUMBER=234XXXXXXXXX

# Company Identity
COMPANY_NAME=OpenSpace
COMPANY_EMAIL=hello@openspace.finance
COMPANY_PHONE=+234 201 3309 599
COMPANY_WEBSITE=https://openspace.finance
COMPANY_TIMEZONE=Africa/Lagos

# Business Operating Hours
BIZ_HOURS_MON_FRI=09:00-17:00
BIZ_HOURS_SAT=CLOSED
BIZ_HOURS_SUN=CLOSED
```